MH1812 Discrete Mathematics
- Continuous Assessment 2

NTU, AY15/16 S2 — 30 March 2016, 09:30AM - 10:30AM @ LT5

Name: Tutorial Group: NTU Email:

Note: ***first 4*** *questions are compulsory. Question 5 is bonus, you get the points if you answer it correctly, you don't lose any points if you leave it or answer wrongly. You can write on the back of paper if there is not enough space. Leave the tutorial group if you can't remember.*

Question 1 (30 points)
Let $A = \{a, b, c, d\}$, and relation R defined on A to be $\{(a, b), (b, c), (c, b), (c, d)\}$,

(a) is R reflexive, symmetric, anti-symmetric, or transitive ? (give explicit justification of your answers)

(b) find R^2, i.e., the composition relation $R \circ R$. (list all elements)

(c) find the transitive closure R^t. (list all elements)

Solution:

(a) R is not reflexive as $(a, a) \notin R$ (can be others)
R is not symmetric as $(a, b) \in R$, $(b, a) \notin R$, and $a \neq b$.
R is not anti-symmetric as $(b, c), (c, b) \in R$ and $b \neq c$.
R is not transitive as $(a, b), (b, c) \in R$ but $(a, c) \notin R$.

(b) $R^2 = \{(a, c), (b, b), (b, d), (c, c)\}$

(c) $R^t = \{(a, b), (a, c), (a, d), (b, b), (b, c), (b, d), (c, b), (c, c), (c, d)\}$

□

Question 2 (25 points) Let A be a set with $|A| = n$ for some positive integer n, and relation R defined on A,

(a) how many possible R are there ? (hint: R is a well defined relation on A if and only if $R \subseteq A \times A$).

(b) how many reflexive R are there ?

(c) how many symmetric R are there ?

Solution:

(a) 2^{n^2}
Let $S = \{(x, y) \mid x \in A \text{ and } y \in A\}$, any subset of S is a relation on A, and there are 2^{n^2} subsets of S.

(b) 2^{n^2-n}
Let $T = \{(x, x) \mid x \in A\}$, for any relation R to be reflexive, $T \subseteq R$, i.e., all elements in T must be contained in R, hence we are looking for R fulfilling the condition: $T \subseteq R \subseteq S$. In other words, $(R - T) \subseteq (S - T)$, there are $n^2 - n$ elements in $S - T$, hence 2^{n^2-n} subsets.

(c) $2^n 2^{\frac{n(n-1)}{2}} = 2^{\frac{n(n+1)}{2}}$.
Let $P = \{(x, y), (y, x) \mid x, y \in A, x \neq y\}$. Note (x, y) and (y, x) appear in pairs, and there are $\binom{n}{2}$ such pairs. Any subset of $T \cup P$ with $(x, y)$and (y, x) appearing in pairs is a symmetric relation, there are $\binom{n}{2} + n$ elements in total, hence $2^{\binom{n}{2}+n} = 2^{\frac{n(n+1)}{2}}$ subsets.

$\square$

Question 3 (20 points) Find an explicit formula for the sequence $a_k = 9a_{k-2}$ for all integers $k \geq 2$, $a_0 = 0$, and $a_1 = 2$.

Solution:

This is a linear homogeneous recurrence with characteristic equation $x^2 = 9$, solving it results in two distinct solutions $x = 3$ and $x = -3$, hence $a_k = u3^k + v(-3)^k$. Substitute $a_0 = u + v = 0$, and $a_1 = 3u - 3v = 2$, we get $u = 3^{-1}, v = -3^{-1}$, hence $a_k = 3^{k-1} + (-3)^{k-1}$. $\square$

Question 4 (25 points) Let $A = \{1, 2, 3, 4\}$, functions $f, g : A \to A$ by the rules $f(x) = 3x \bmod 5$ and $g(x) = 2x \bmod 5$,

(a) find $f(x)$ for all $x \in A$, and determine whether f is onto or one-to-one.

(b) find the formula of $g \circ f$.

(c) is $f^{-1} = g$?

Solution:

(a) $f(1) = 3, f(2) = 1, f(3) = 4, f(4) = 2$, every value in the co-domain A appears exactly once, hence f is both onto and one-to-one

(b) $g \circ f(x) = x \mod 5$.

(c) $g \circ f$ is the identity function, hence $f^{-1} = g$.

$\square$

Question 5 (Bonus Question) (10 points)
For any real number $x \in \mathbb{R}$, $\lfloor x \rfloor$ denotes an integer $z \in \mathbb{Z}$ such that $z \leq x < z+1$, i.e., the biggest integer less than or equals to x. Find a formula for

$$\sum_{k=1}^{n^2} \lfloor \sqrt{k} \rfloor$$

for some non-negative integer k and n. (You may find $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ and $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ useful)

Solution: $\sum_{k=1}^{n^2} \lfloor \sqrt{k} \rfloor = n + \sum_{i=1}^{n-1} \sum_{k=i^2}^{(i+1)^2-1} \lfloor \sqrt{k} \rfloor = n + \sum_{i=1}^{n-1} \sum_{k=i^2}^{(i+1)^2-1} i = n + \sum_{i=1}^{n-1} (2i+1) \cdot i = n + \sum_{i=1}^{n-1} 2i^2 + \sum_{i=1}^{n-1} i = n + \frac{(n-1)n(2n-1)}{3} + \frac{(n-1)n}{2}$. $\square$